

PCOS Pathfinder

A Tiered, Explainable Decision-Support Prototype for Earlier PCOS Detection and Differential Diagnosis

Biohackathon 2026: Women's Health

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Abstract

PCOS Pathfinder is a data-grounded clinical decision-support prototype for earlier identification of Polycystic Ovary Syndrome (PCOS) and broader differential workup in women's health. The system anchors on the required PCOS clinical dataset (n=541, Kerala, India), layers a synthetic endometriosis-overlap module for differential reasoning, and uses a lightweight single-cell gene-list check as biological rationale. The delivered artifacts are six reproducible notebooks, three saved scikit-learn models, a Streamlit clinician interface, exported metrics and figures, and this report. Thresholds are selected from out-of-fold training probabilities (no test leakage), risk tiers are anchored to each model's action threshold, and patient-level SHAP attributions are provided for both PCOS models. On the held-out test split the enhanced model reaches ROC-AUC 0.953 with recall 0.886 and specificity 0.902. The system is framed throughout as decision support, not standalone diagnosis.

Headline metric	Screening	Enhanced
ROC-AUC	0.896	0.953
Recall (sensitivity)	0.886	0.886
Specificity	0.685	0.902
NPV (rule-out value)	0.926	0.943
F2 score	0.799	0.871

Held-out test split (n=136) at CV-selected high-recall thresholds. F1 is deliberately suppressed by the high-recall threshold — see §6.2.

1. Challenge Context and Project Goal

The Biohackathon 2026 challenge asks teams to improve diagnostic accuracy for women's health conditions using available data and a feasible implementation pathway. PCOS is the core case because it is common, heterogeneous, and frequently diagnosed late: clinicians must synthesise menstrual, metabolic, hormonal, ovarian, dermatologic, and reproductive information, and overlapping conditions are routinely missed.

A defensible solution therefore needs to do more than predict a binary label. It should help clinicians identify high-risk patients earlier, explain the drivers of risk in clinically familiar terms, and avoid tunnel vision when adjacent conditions share presentations. The project goal is to build **PCOS Pathfinder**, a clinician-facing tiered decision-support workflow with three linked components:

- **Frontline screening model** — high-recall PCOS risk flag using symptoms and basic vitals only, intended for primary care, telehealth, and community clinics.
- **Enhanced diagnostic-support model** — a stronger model that uses laboratory and ultrasound features once further investigations are available.
- **Endometriosis-overlap prompt** — a synthetic-data workflow nudge that prompts broader differential workup when symptoms overlap. *Not* a diagnostic claim.

2. Data Sources

Dataset	Rows / samples	Use
PCOS clinical (Kerala, India)	541 patients	Required modelling foundation
Endometriosis (synthetic)	10,000 rows	Differential-overlap prompt
PCOS single-cell archive	20 samples	Biological rationale (gene-list peek)
Endometrium single-cell archive	2 samples	Biological context only

The PCOS workbook's modelling sheet (Full_new) contains 44 columns and a binary target PCOS (Y/N). The class split is 364 non-PCOS to 177 PCOS, a roughly 2:1 imbalance.

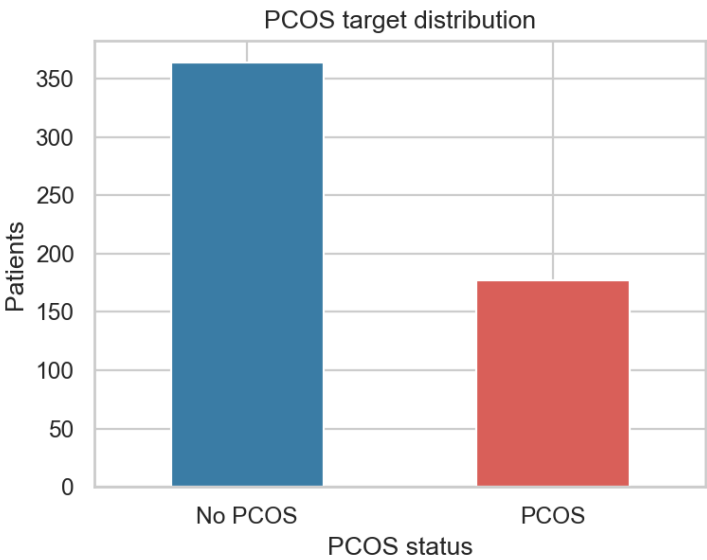


Figure 1. PCOS class distribution in the required dataset.

The supplementary endometriosis dataset has seven variables: age, menstrual irregularity, chronic pain level, hormone-level abnormality, infertility, BMI, and diagnosis. Because it is synthetic, the resulting model is treated as a workflow prompt only. The single-cell archives are 10x-style outputs; this project deliberately limits itself to a gene-list peek rather than a full single-cell analysis, because the judging weights reward clinical usefulness over single-cell technique depth.

3. Data Cleaning and Quality Controls

Cleaning lives in the notebook setup cells generated from the deterministic regenerator script (scripts/■create_■training_■notebooks.mjs). The key steps:

- Standardise column names to lowercase snake case; coerce numerics and log every non-numeric cell.
- Drop identifiers (sl_no, patient_file_no).
- Drop blood_group (codes 11–18 are not ordinal) and marriage_status_yrs (clinically sensitive, no diagnostic signal).
- Engineer cycle_irregular_flag from cycle_r_i (Kaggle coding: 2 = regular, 4/5 = irregular).
- Cap visibly-impossible outliers in FSH, LH, FSH/LH, vitamin D3, progesterone.
- Use median imputation *inside* each scikit-learn pipeline so the rules travel with the saved artifact.

3.1 Data Caveat: "Cycle Length" Is Actually Bleeding Duration

The source column labelled **Cycle length(days)** ranges 0–12 with a median of 5 across both classes — this is the *duration of menses bleeding in days*, not the menstrual cycle interval. The Kaggle dataset documentation is misleading on this point. The Streamlit prototype surfaces the field with the corrected label and a 0–12 range so clinicians do not accidentally enter a typical 28-day cycle value, which would be out-of-distribution and silently produce wrong predictions.

3.2 Logged Coercions

Column	Example value	Count
ii_beta_hcg_miu_ml	1.99.	1
amh_ng_ml	a	1

4. Model Design

4.1 Frontline Screening Model

Intended for primary care, telehealth, and community settings. Lifestyle proxies (fast_food, reg_exercise) are intentionally excluded — bias-prone, hard to validate at intake, and likely to encode confounders rather than physiology.

Screening feature set (13 features)

Age, BMI, cycle pattern (raw and engineered flag), duration of menses bleeding, weight gain, hair growth, skin darkening, hair loss, pimples, random blood sugar, BP systolic, BP diastolic.

4.2 Enhanced Diagnostic-Support Model

Enhanced model additional features (14 added, 27 total)

Haemoglobin, FSH, LH, FSH/LH ratio, TSH, AMH, prolactin, vitamin D3, progesterone, left and right follicle counts, left and right follicle sizes, endometrium thickness.

4.3 Candidate Algorithms

Both PCOS notebooks compare four models: a most-frequent dummy baseline, logistic regression with class-weight balancing and standard scaling, a random forest with class-weight balancing, and gradient boosting. Selection is by 5-fold stratified cross-validation on the training split, ranking by ROC-AUC and breaking ties on recall. Across runs, random forest wins both contests.

5. Validation Strategy

A single stratified train/test split with `test_size=0.25` and `random_state=42`. Model selection uses 5-fold stratified cross-validation on the *training* split. For each chosen model, action thresholds are selected using out-of-fold training probabilities from `cross_val_predict(method="predict_proba")`. The threshold sweep targets recall ≥ 0.90 on the training out-of-fold predictions while maximising specificity, then precision, then F1.

The held-out test set is then evaluated **exactly once**. This eliminates the earlier risk of choosing a threshold on the test set and reporting optimistic metrics.

6. Results

6.1 Held-Out Performance at the Chosen Thresholds

Model	Thresh.	ROC-AUC	Recall	Spec.	PPV	NPV	F2
Screening	0.285	0.896	0.886	0.685	0.574	0.926	0.799
Enhanced	0.380	0.953	0.886	0.902	0.812	0.943	0.871
Endo (synth.)	0.510	0.660	0.628	0.618	0.531	—	0.609

The enhanced model improves specificity ($0.685 \rightarrow 0.902$) and precision ($0.574 \rightarrow 0.812$) while preserving the same recall as the screening model. NPV is 0.926 for screening and 0.943 for enhanced — when either model says "no PCOS", it is right >92% of the time, which is the right framing for ruling-out triage.

6.2 Why F1 Is Not the Right Headline Metric for Screening

The screening F1 of 0.696 is intentionally suppressed by the high-recall threshold. F1 weights precision and recall equally, but in primary-care screening the cost of a missed PCOS case (years of metabolic damage, infertility risk) is much larger than the cost of an over-referral (one additional clinical visit). Two more screening-aligned metrics behave very differently:

- **Recall (0.886)** — the screening tool catches 39 of the 44 PCOS-positive patients in the held-out cohort.
- **NPV (0.926)** — when the tool says "no PCOS", the clinician can trust it.
- **F2 (0.799)** — weights recall four times more than precision; the screening-friendly counterpart to F1.

Table 4 compares the chosen high-recall threshold to a balanced 0.50 threshold for each model. The screening row makes the trade-off most visible: the balanced threshold raises F1 from 0.70 to 0.81 but loses four PCOS cases (recall $0.886 \rightarrow 0.795$). At the high-recall threshold, F2 is 0.80; at the balanced threshold, F2 is also 0.80 — F2 is largely indifferent, so the high-recall threshold is the right pick under a

sensitivity-first cost function.

Model / threshold	Recall	Spec.	PPV	NPV	F1	F2	TN, FP, FN, TP
Screening, t=0.285 (chosen)	0.886	0.685	0.574	0.926	0.696	0.799	63, 29, 5, 39
Screening, t=0.500 (balanced)	0.795	0.913	0.814	0.903	0.805	0.799	84, 8, 9, 35
Enhanced, t=0.380 (chosen)	0.886	0.902	0.812	0.943	0.848	0.871	83, 9, 5, 39
Enhanced, t=0.500 (balanced)	0.818	0.957	0.900	0.917	0.857	0.833	88, 4, 8, 36

Table 4. Threshold trade-off on the held-out test set.

For the enhanced model the high-recall threshold actually *dominates* the balanced threshold under F2 (0.871 vs 0.833), so there is no headline F-score loss from the screening-friendly choice.

6.3 ROC Curves and Confusion Matrices

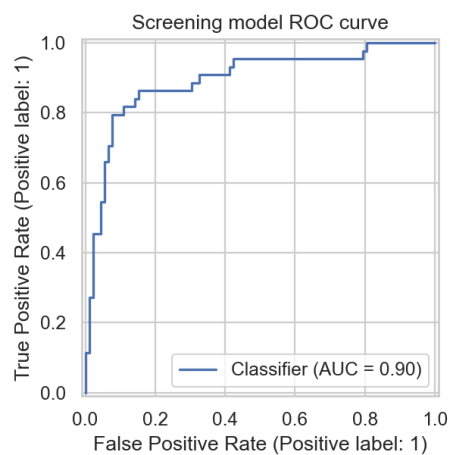


Figure 2. Screening ROC, AUC = 0.896.

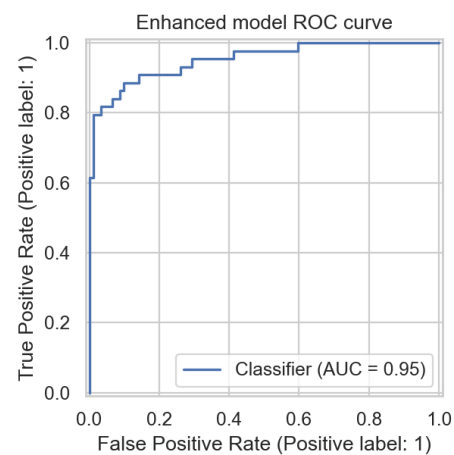


Figure 3. Enhanced ROC, AUC = 0.953.

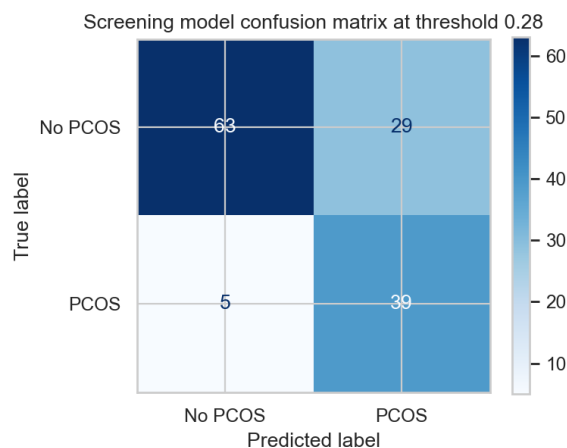


Figure 4. Screening confusion at t = 0.285.

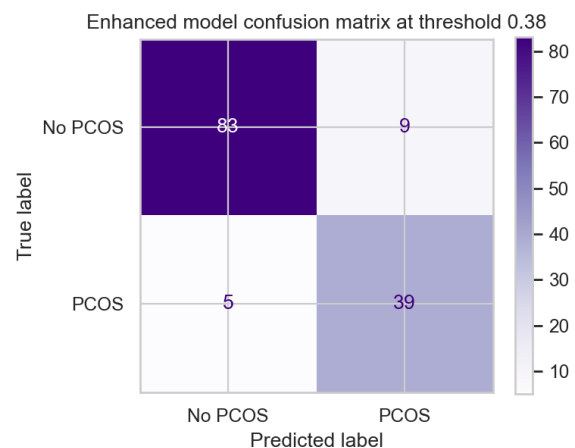


Figure 5. Enhanced confusion at t = 0.380.

7. Explainability

Explainability is treated as a core product feature, not an afterthought.

7.1 Global Drivers

Global feature importances are exported per model. The screening model leans on symptom presentation (skin darkening, hair growth, weight gain) and basic vitals; the enhanced model is dominated by ultrasound (follicle counts) once those features are available.

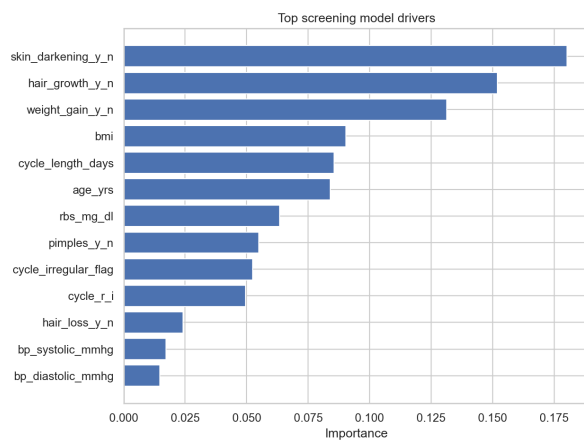


Figure 6. Screening global drivers.

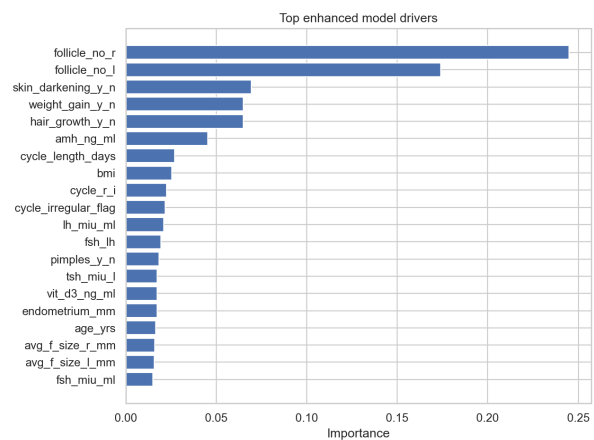


Figure 7. Enhanced global drivers.

7.2 Patient-Level SHAP Attributions

For tree models, `shap.TreeExplainer` produces fast and exact attributions. For logistic-regression fall-backs, each model artifact ships with a 100-row transformed training background that `shap.LinearExplainer` consumes. Demo cases are drawn from the held-out test slice (reproduced from the same random seed), so the explanations describe rows the model never saw during training.

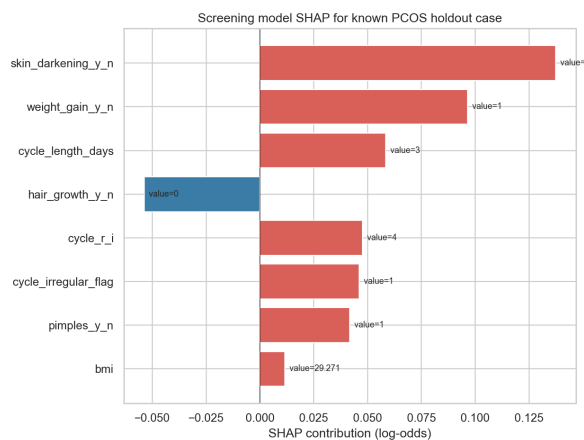


Figure 8. Screening SHAP, holdout PCOS-positive case.

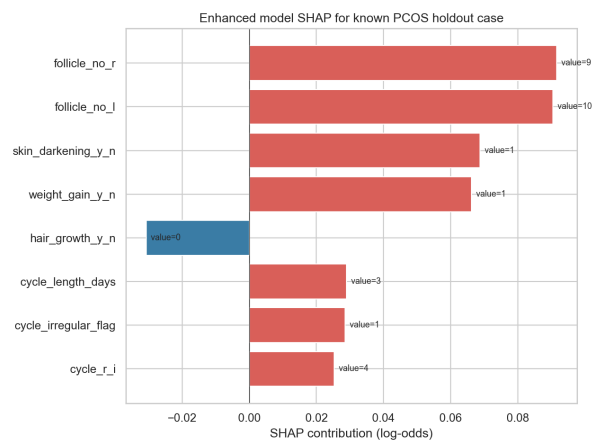


Figure 9. Enhanced SHAP, holdout PCOS-positive case.

8. Differential-Diagnosis Prompt

The endometriosis-overlap model is intentionally modest. It is trained on the supplementary synthetic dataset (10,000 rows) using six features (age, BMI, menstrual irregularity, chronic pain level, hormone-level abnormality, infertility) and reaches ROC-AUC 0.66 on a held-out split — a level that reflects the limits of the underlying synthetic data rather than a modelling failure. In the prototype, a positive overlap score raises a warning that the clinician should consider a broader gynaecologic differential workup, especially when chronic pain and infertility are reported alongside menstrual irregularity. It is *not* described as a validated endometriosis diagnostic model.

9. Prototype

The Streamlit prototype at `src/app.py` loads the three joblib model artifacts and presents three tabs:

- **Patient assessment** — intake form with collapsible sections for labs/ultrasound and differential-workup inputs, followed by stacked screening, enhanced, and endometriosis-overlap results. Each PCOS result shows probability, action threshold, threshold-anchored risk tier, and top SHAP drivers.
- **Population view** — symptom prevalence by PCOS status (source cohort, n=541) and headline model metrics.
- **About** — plain-language model descriptions, threshold strategy, and caveats.

Risk tiers are anchored to each model's threshold so the displayed tier never contradicts the action flag:

- **Low**: probability below threshold.
- **Moderate**: probability from threshold to threshold + 0.20 (capped at 0.90).
- **High**: probability above the moderate band.

10. Biological Rationale from Single-Cell Feature Lists

Notebook 06 extracts feature symbols from one PCOS single-cell sample and checks for the presence of a curated list of literature-supported PCOS genes across steroidogenesis (CYP17A1, CYP19A1, HSD3B2, STAR), androgen signalling (AR, SRD5A1, SRD5A2, HSD17B3), follicular development (FSHR, LHCGR, GATA4), AMH signalling (AMH, AMHR2), insulin signalling (INSR, IRS1, IRS2, INS), and PCOS GWAS loci (DENND1A, THADA, FSHB). All 20 curated genes appear in the per-sample feature lists, giving the slide deck a defensible biological rationale for the clinical features the model uses. No QC, normalisation, integration, clustering, or differential expression analysis is claimed.

11. Limitations and Risk Controls

- **Small, single-source PCOS dataset.** 541 patients from 10 hospitals in Kerala, India is not enough for deployment-level validation, and the geography may not generalise.
- **Synthetic endometriosis data.** The differential module is a workflow prompt, not a diagnostic claim.
- **Equity variables are unavailable.** Race, income, geography, and access-to-care measures are missing, so equity claims rest on workflow design and low-resource feasibility rather than measured outcome disparities.
- **Clinical framing.** All outputs are framed as risk, triage, and decision support, not diagnosis.
- **Version sensitivity.** scikit-learn model artifacts are sensitive to version skew; `requirements.txt` pins the exact versions used during training.
- **Threshold philosophy.** Tiered thresholds favour sensitivity; clinicians who prefer specificity-first triage can switch to the balanced threshold reported in Table 4 without retraining.

12. Reproducible Artifacts

Artifact	Purpose
notebooks/01_pcos_data_audit_and_eda.ipynb	Loading, cleaning, audit, EDA, processed CSV export.
notebooks/02_train_pcos_screening_model.ipynb	Training, threshold selection, evaluation: screening.
notebooks/03_train_pcos_enhanced_model.ipynb	Training and evaluation: labs/ultrasound model.
notebooks/04_train_endometriosis_overlap_model.ipynb	Synthetic-data differential prompt model.
notebooks/05_thresholds_explainability_and_demo.ipynb	SHAP, held-out demo cases, model card.
notebooks/06_single_cell_gene_peek.ipynb	Lightweight biological-rationale gene-list check.
src/app.py	Streamlit clinician-facing prototype.
outputs/models/*.joblib	Saved model artifacts (with SHAP backgrounds).
outputs/metrics/*	Held-out metrics, threshold trade-offs, SHAP tables, model card.
scripts/create_training_notebooks.mjs	Deterministic notebook regenerator (source of truth).
report/build_pdf.py	Reportlab generator for this PDF.
PROJECT_PLAN.md, README.md, SUBMISSION.md	Plan, quickstart, packaging checklist.

13. Conclusion

The defensible pitch for PCOS Pathfinder is not automated diagnosis. It is a practical, explainable, tiered workflow that helps clinicians identify high-risk patients earlier (recall ≥ 0.88), choose appropriate next investigations (the enhanced model dominates F2 at the same recall), and avoid overlooking adjacent women's health conditions (the differential prompt). The cleaning rules, threshold strategy, and SHAP attributions are reproducible end-to-end from the six numbered notebooks; the Streamlit prototype loads the same artifacts a clinician would use; and the limitations are stated up-front in the model card.